# Interpretable Analysis of Correlated Knowledge Gaps in GAN-Generated Images

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Computer Vision and Pattern Recognition [Section: B]

***Abstract*— Generative Adversarial Networks (GANs) can produce highly realistic images, often making it difficult to distinguish from real ones for human observers. However, previous studies have shown that machine learning classifiers can reliably detect GAN-generated images indicating the presence of systematic artifacts. Recent work has introduced the concept of "knowledge gaps" referring to these artifacts that are even correlated across independently trained GAN instances. In this work, we present an interpretable analysis of correlated GAN knowledge gaps using both spatial and frequency-domain perspectives. Experiments are conducted on the MNIST dataset using multiple independently trained DCGAN models. A binary CNN classifier is trained to distinguish real images from GAN-generated images. To understand the classifier's decision behaviour, Grad-CAM is applied to visualize regions that influence classification decisions. Additionally, FFT-based frequency analysis is used to examine spectral differences between real and generated images. The results show that the classifier relies on non-semantic artifacts rather than digit identity. Grad-CAM visualizations and cosine similarity measurements demonstrate that these artifacts are highly correlated across independently trained GAN instances. Frequency-domain analysis further reveals consistent spectral irregularities shared across GANs. These findings provide interpretable evidence of correlated knowledge gaps in GAN-generated images.***

***Keywords*— GANs, Artifacts, Knowledge Gaps, Grad-CAM, Frequency-Domain Analysis**

## I. Introduction

Generative Adversarial Networks (GANs) have achieved remarkable success in synthesizing realistic images across a wide range of visual domains. Since their introduction by Goodfellow et al. [1], GAN-based models have been applied to image generation tasks where generated images often appear indistinguishable from real images to human observers.

Despite their visual realism, several studies have demonstrated that GAN-generated images can be reliably detected using machine learning-based classifiers [2],[3]. This observation suggests that GANs introduce systematic artifacts that distinguish generated images from real data. Recent work by Pathak and Dufour [4] introduced the concept of "knowledge gaps", referring to the consistent presence of artifacts across GAN-generated samples. They also revealed that such knowledge gaps are correlated. This finding challenges the assumption that GAN artifacts are instance-specific. While prior work has established the existence of these knowledge gaps, most studies focus primarily on classifier accuracy and detection performance [2],[4], rather than interpretability. In this work, we aim to bridge this gap by providing an interpretable analysis of GAN knowledge gaps using both spatial and frequency-domain perspectives. We apply Gradient-weighted Class Activation Mapping (Grad-CAM) [5] for identification of image regions that influence classification decisions. Additionally, we perform FFT-based frequency analysis to examine spectral differences.

## II. Literature Review

Research on Generative Adversarial Networks (GANs) has rapidly evolved since their introduction. Early GAN-based models demonstrated the feasibility of generating realistic images, but many studies revealed that such images often contain artifacts that distinguish them from real data [1].

A large body of work investigated the detection of GAN-generated images using convolutional neural networks. Wang et al. [2] showed that CNN-based classifier can detect GAN-generated images with surprisingly high accuracy. This finding suggested that GANs leave systematic traces that can be learned by the classifiers.

Several works have explored the nature of these artifacts. Yu et al. [6] and Marra et al. [7] demonstrated that they contain architecture specific and instance specific traces.

More recently, Pathak and Dufour [4] introduced a novel perspective by systematically studying GAN. Their work demonstrated that artifacts are correlated across independently trained GAN instances. Their work provided strong evidence that GAN artifacts are structured and repeatable rather than random.

Despite these advances, much of the existing literature focuses on detection accuracy and generalization, with limited focus on interpretability. Some studies showed visualization techniques or frequency analysis to highlight differences between real and fake images [8],[9]. But few works explicitly analyse which image regions classifiers attend to and whether such attention patterns are consistent across different GAN instances.

In contrast to prior detection focused approaches, our work aims to provide an interpretable analysis of GAN knowledge gaps. By combining Grad-CAM based spatial analysis with frequency domain analysis, this study seeks to explain what visual and spectral features make these knowledge gaps and how they are shared across independently trained GANs.

## III. METHODOLOGY

This section describes the dataset, model architecture, training procedure and analysis techniques used to investigate the visual and frequency-domain characteristics of correlated GAN "knowledge gaps."

### *A. Dataset*

All experiments were conducted using MNIST handwritten digit dataset, which consists of 60000 training images and 10000 test images. Each image is a size of 28×28 pixels in grayscale. It was selected for its simplicity and suitability for interpretability-focused analysis.

### *B. GAN Architecture and Training*

To generate synthetic images, we employed a Deep Convolutional Generative Adversarial Network (DCGAN) architecture. The generator consists of a series of fully connected and transposed convolutional layers. It upsamples a 100-dimensional latent noise vector into a 28×28 grayscale image. Batch normalization and LeakyReLU activations were used to stabilize training. The discriminator is a convolutional neural network composed of strided convolutional layers followed by fully connected layers.

Three independent GAN instances (GAN0, GAN1, GAN2) were trained using identical architectures and training data but with random initializations. Each GAN was trained for 5 epochs using the Adam optimizer.

### *C. Classifier Architecture and Training*

To detect GAN-generated images, a binary convolutional neural network (CNN) classifier was trained. It consists of two convolutional layers with ReLU activations and max pooling, followed by fully connected layers and a single output neuron representing the real or fake decision. It was trained on a balanced dataset of 40000 images, including 20000 real images from MNIST and 20000 fake images generated by GAN0 and GAN1.

### *D. Grad-CAM Based Interpretability Analysis*

To identify which image regions the classifier relied on for its decisions, Grad-CAM was applied to the trained classifier. It produces heatmaps by computing the gradient of the classifier's output. To quantify consistency across GAN instances, mean Grad-CAM heatmaps were computed.

### *E. Cross-GAN Correlation Analysis*

To measure the similarity of classifier attention across different GAN instances, cosine similarity was computed. High similarity values indicate that the classifier focuses on similar regions when evaluating images, suggesting correlated artifact patterns. This analysis provides a quantitative measure of whether GAN "knowledge gaps" are shared across independently trained generators.

### *F. Frequency-Domain Analysis*

Frequency-domain analysis was conducted using the Fourier Transform (FFT). The power spectrum of each image was computed to examine how energy is distributed across spatial frequencies.

All models were implemented using TensorFlow and trained using Google Colab. The same preprocessing, training and evaluation pipeline was applied consistently across all experiment to ensure fair comparison.

**The complete implementation and experimental code for this work are publicly available at: https://github.com/Hossain-Arafat/cvpr/blob/main/FINAL/Paper/gan_artifact_analysis.ipynb**

## IV. EXPERIMENT AND RESULT

### *A. Experimental setup*

We conducted experiments on the MNIST dataset, consisting of grayscale images of handwritten digits. Image size 28×28 and normalized to the range [-1,1]. Three DCGAN generators (GAN0, GAN1, GAN2) were trained for 5 epochs using the same training data.

To study the "knowledge gaps" we trained a binary CNN classifier to distinguish between real MNIST images and GAN generated images. The classifier was trained on 40000 images where 20000 images were real

and 20000 images were fake. Fake samples were taken from GAN0 and GAN1 only while GAN2 was kept unseen during classifier training.

For analysis, we applied:

- Grad-CAM to visualize classifier attention
- Cosine similarity between mean Grad-CAM heatmaps across GANs
- Frequency-domain (FFT) analysis to study spectral artifacts

### *B. GAN Training Behaviour and Visual Quality*

Across all three GAN instances, early epoch (epoch 1) showed noisy pattern while later epoch (epoch 5) showed structured digit strokes. Figure 1 shows representative samples generated by GAN instances:

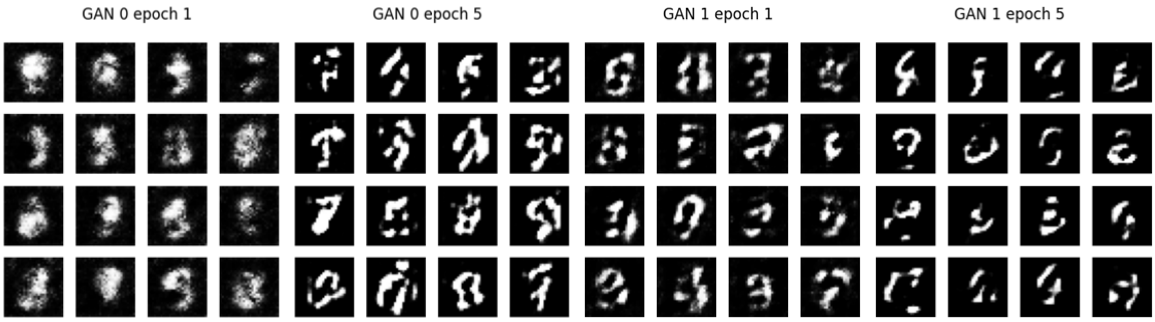


Fig. 1 Samples generated by GAN instances

Importantly, although the GANs were trained independently, their final outputs showed similar types of imperfections. This suggests that the generator converged toward correlated artifact patterns rather than random errors. This observation aligns with the hypothesis that GANs trained under identical settings tend to cover similar regions of artifact.

### *C. Classifier Performance*

The CNN classifier achieved very high-performance during training and validation:

- Final training accuracy: 99.92%
- Final validation accuracy: 99.77%
- Validation loss remained consistently low

It means the classifier learned discriminative features that can distinguish real images from GAN generated images. A high performing classifier is essential for reliable interpretability analysis.

### *D. Grad-CAM Analysis of Knowledge Gaps*

To investigate what features the classifier relies on, Grad-CAM was applied to both real and fake images. Representative Grad-CAM visualizations are shown below:

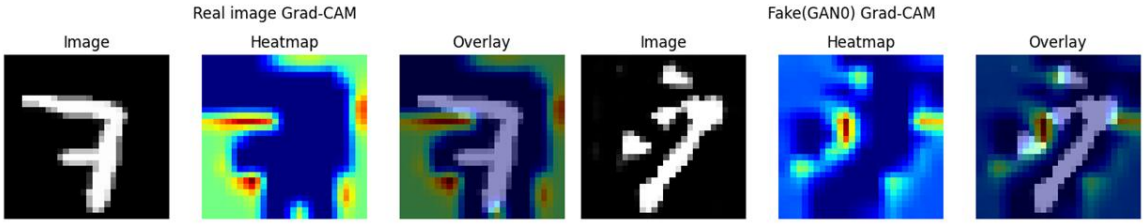


Fig. 2 Grad-CAM visualizations of real image(left) and fake image(right)

For real image, Grad-CAM heatmaps primarily highlighted:

- Digit stroke boundaries
- Semantically meaningful regions of the digit
- Minimal activation in background regions

This indicates that for real images, the classifier attends to human-interpretable, semantic features.

In contrast, Grad-CAM heatmaps for GAN-generated images focused on:

- Irregular stroke transitions
- Noisy and fragmented regions
- Non-semantic background artifacts

These highlighted areas reflect slight errors introduced by GAN rather than digit identity. This qualitative distinction implies that GAN classifiers take advantage of non-semantic visual phenomena, which make up the "knowledge gaps" discussed in prior work.

### *E. Cross-Gan Correlation of Grad-CAM Heatmaps*

To identify whether these artifacts are shared across GAN instances, we computed the mean Grad-CAM heatmap for each GAN using 200 generated samples and measured pairwise cosine similarity.

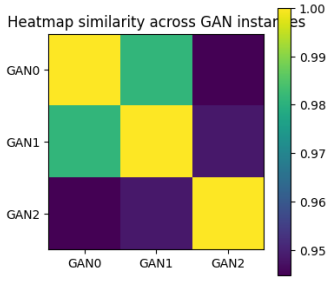


Fig. 3 Mean Grad-CAM heatmap

The resulting similarity matrix shows consistently high values:
GAN0-GAN1: 0.98
GAN0-GAN2: 0.94
GAN1-GAN2: 0.95

These high similarity scores (0.94-0.98) indicate that independently trained GANs show highly correlated classifier attention patterns, even when one GAN (GAN2) was unseen during classifier training.

This outcome offers a compelling quantitative proof that GAN "knowledge gaps" are shared structural artifacts brought about by the generator architecture and training process rather than being instance specific.

### *F. Frequency-Domain Analysis*

To further analyse the nature of these shared artifacts, we examined the frequency-domain characteristics of real and fake images using Fourier Transform (FFT).

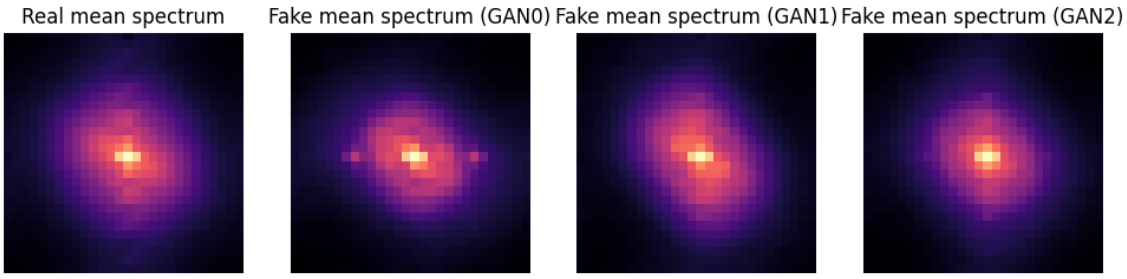


Fig. 4 Mean power spectra visualization

Mean power spectra visualizations reveal clear differences:

- Real images show smoother, more evenly distributed frequency energy.
- GAN-generated images showed stronger concentration in specific frequency bands, particularly in high frequency regions

Moreover, the mean spectra of GAN0, GAN1 and GAN2 are visually similar, meaning that independently trained GANs introduced consistent frequency-domain artifacts.

These findings suggest that GAN classifier may partially rely on frequency-based cues. It largely remains invisible to human observers but systematically present across GAN instances

## V. Discussion and Limitations

The experimental result shows clear evidence that GAN classifiers rely on non-semantic artifacts rather than digit identity. Through Grad-CAM analysis, we observed that for real images, the classifier attended to meaningful digit strokes and boundaries. But for GAN-generated images, the attention shifted toward irregular stroke transitions, fragmented regions and background noise. These regions are largely not visible to human observers but consistently exploited by the classifier.

A key finding of this work is the strong correlation of attention across independently trained GAN instances. Despite being trained independently, GAN instances produced artifacts that resulted in cosine similarity values exceeding 0.94 across all GAN pairs. This means that the observed "knowledge gaps" are not instance-specific rather arise from shared properties of generator architecture and training process.

The frequency-domain analysis further supports this conclusion. GAN-generated images showed consistent spectral irregularities, particularly in higher-frequency regions. But real images showed smoother and more evenly distributed frequency energy.

These findings are consistent with prior work that reports correlated "knowledge gaps" in GANs. Importantly, the results also suggests that such artifacts generalize to GAN instances which are unseen during classifier training.

This study has several limitations. First, experiments were conducted only on the MNIST dataset, which is low-dimensional and relatively simple compared to real world image dataset. Second, we used basic DCGAN and trained for a limited number of epochs. More advanced architecture, such as StyleGAN may produce different or more reliable artifact patterns. Third, only a single classifier architecture was evaluated. Finally, the analysis focused only on visual and frequency-domain features. Future work could extent this analysis to higher-resolution datasets, multiple advanced GAN and classifier settings.

## VI. Conclusions

In this work, we presented an interpretable analysis of knowledge gaps by examining the spatial and frequency-domain features. We demonstrated that classifiers rely primarily on non-semantic artifacts rather than digit identity. Through Grad-CAM analysis, we showed that classifier attention consistently focuses on irregular stroke transitions, fragmented regions and background noise in GAN-generated images. Cosine similarity further revealed that these attention patterns are highly correlated across independently trained GAN instances. Moreover, frequency-domain analysis confirmed the presence of shared spectral irregularities in GAN generated images. Together, these findings provide clear and interpretable evidence that GAN knowledge gaps arise from generator architecture and training processes. Although this study was conducted on simplified dataset and model, it offers valuable insights into what GAN classifiers learn and why detection generalizes across generators. This work serves as a foundation for future studies on more complex datasets and architecture.


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## AUTHOR CONTRIBUTIONS

The author, Arafat Hossain (ID: 23-50277-1) solely contributed to all aspects of this work, including problem formulation, literature review, model implementation, experimental design, results analysis and manuscript writing.

## References


[1] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 27, 2014.

[2] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, "CNN-generated images are surprisingly easy to spot… for now," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 8695–8704.

[3] D. Gragnaniello, D. Cozzolino, F. Marra, G. Poggi, and L. Verdoliva, "Are GAN generated images easy to detect? A critical analysis of the state-of-the-art," in *Proceedings of the IEEE International Conference on Multimedia and Expo (ICME)*, 2021, pp. 1–6.

[4] A. Pathak and N. Dufour, "Sequential training of GANs against GAN-classifiers reveals correlated knowledge gaps present among independently trained GAN instances," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[5] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[6] N. Yu, L. S. Davis, and M. Fritz, "Attributing fake images to GANs: Learning and analyzing GAN fingerprints," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 7556–7566.

[7] F. Marra, D. Gragnaniello, L. Verdoliva, and G. Poggi, "Do GANs leave artificial fingerprints?" in *Proceedings of the IEEE Conference on Multimedia Information Processing and Retrieval (MIPR)*, 2019, pp. 506–511.

[8] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, D. Kolossa, and T. Holz, "Leveraging frequency analysis for deep fake image recognition," in *Proceedings of the International Conference on Machine Learning (ICML)*, 2020, pp. 3247–3258.

[9] D. Cozzolino, D. Gragnaniello, G. Poggi, and L. Verdoliva, "Towards universal GAN image detection," in *Proceedings of the International Conference on Visual Communications and Image Processing (VCIP)*, 2021, pp. 1–5.